ChinaBay Backend Plan

Database diagram + build plan before backend work

Single vendor (vendor-ready for future marketplace)

Default language: Bangla - LTR; Arabic - RTL supported

32 tables - Laravel 13 / PHP 8.4 / MySQL

1. Module to table map

Checkout

Guest checkout, address + district/area delivery charge, COD + online payment, summary, coupon, confirmation.

Orders

Placed > Confirmed > Processing > Shipped > Out for Delivery > Delivered, tracking, cancel/return/exchange, history.

Customer account

Profile, saved addresses, order history, wishlist, reviews, coupons, loyalty, notifications.

Admin

Sales dashboard, pending orders, low stock, best sellers, product/stock/variants/media/pricing control.

Marketing & analytics

Coupons, flash sales, campaigns, abandoned cart, recommendations, recently viewed, referral/loyalty, Pixel/Analytics.

AI features

Title/description copy (bn+en), SEO keywords, captions, image tools, chatbot, recommendations, sales analysis, voice-to-text.

Delivery & payment

COD, district/thana/area charges, courier API, bKash/Nagad/SSLCommerz, Messenger/WhatsApp, OTP/phone verify.

Security

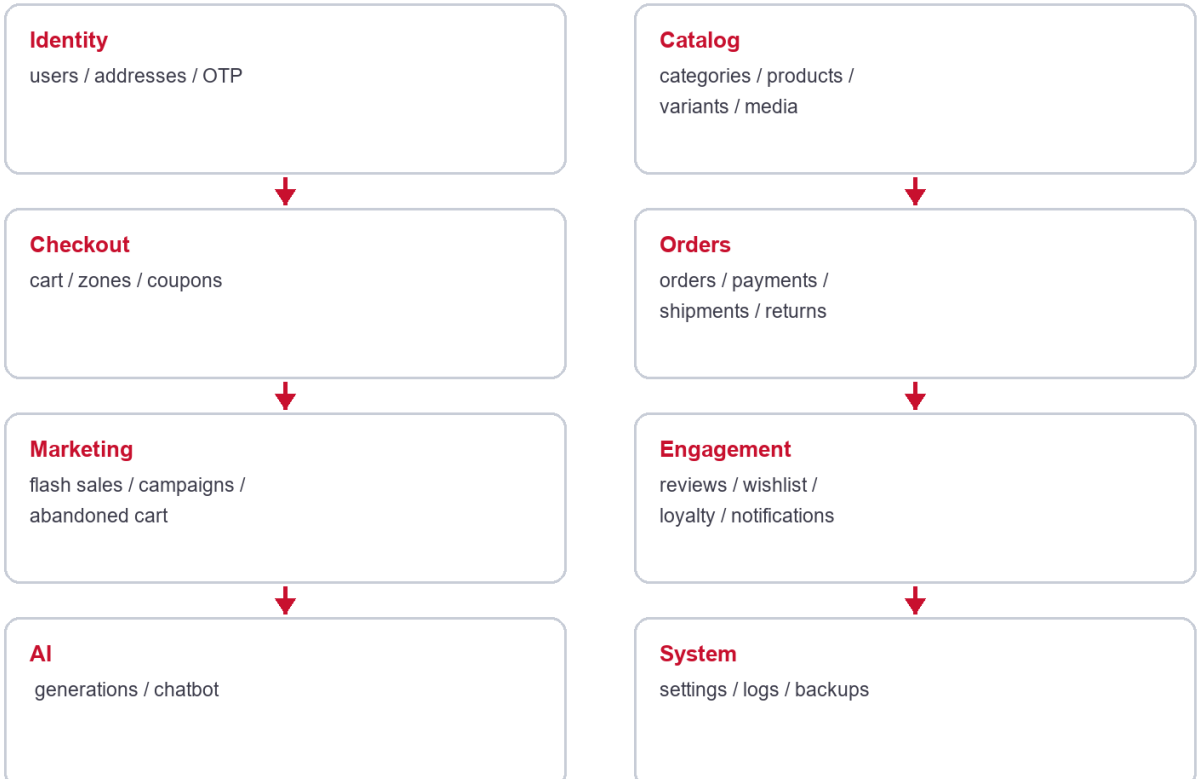
HTTPS, secure pay, OTP, admin 2FA, roles, backups, fraud controls.

Diagram pages

- Module overview
- Identity, account and verification
- Catalog and inventory
- Cart, checkout, delivery and payment setup
- Orders, money and fulfillment
- Campaigns, recovery and engagement
- AI features and system tables

2. Module overview

ChinaBay module map (single vendor, vendor-ready)



Identity, account and verification

Identity, account and verification

users

[PK] id	bigint
name	varchar(100)
[UQ] phone	varchar(20)
email	varchar(190)
password	varchar(255)
role	enum
avatar	varchar(255)
locale	char(2)
loyalty_points	int
+4 more (see spec)	

addresses

[PK] id	bigint
[FK] user_id	bigint
label	varchar(30)
recipient/phone	varchar
district/thana/area	varchar
street/postal	varchar
is_default	bool

phone_otps

[PK] id	bigint
phone	varchar(20)
code_hash	varchar(255)
purpose	enum
attempts	int
expires_at	ts
consumed_at	ts?

Catalog and inventory

Catalog and inventory

categories

[PK] id	bigint
parent_id	bigint
name_en/bn	varchar
[UQ] slug	varchar
icon/image	varchar?
sort_order	int
is_active	bool

brands

[PK] id	bigint
[UQ] name/slug	varchar
logo	varchar?
is_active	bool

products

[PK] id	bigint
vendor_id	bigint?
[FK] category_id	bigint
[FK] brand_id	bigint?
[UQ] name/slug	varchar
subtitle	varchar?
desc_en/bn	text?
price/old_price/cost	decimal
[UQ] sku	varchar

+4 more (see spec)

product_variants

[PK] id	bigint
[FK] product_id	bigint
[UQ] name/sku	varchar
price_override	decimal?
stock_qty	int
attributes	json

product_media

[PK] id	bigint
[FK] product_id	bigint
[FK] variant_id	bigint?
type	enum
url	varchar(500)
sort/is_primary	int/bool

Cart, checkout, delivery and payment setup

Cart, checkout, delivery and payment setup

delivery_zones

[PK] id	bigint
district	varchar
thana/area	varchar?
charge	decimal
[FK] courier_id	bigint?
eta_days	int?
is_active	bool

couriers

[PK] id	bigint
[UQ] name/code	varchar
api_config	json?
is_active	bool

carts

[PK] id	bigint
[FK] user_id	bigint?
session_id	varchar?
[FK] coupon_id	bigint?

cart_items

[PK] id	bigint
[FK] cart_id	bigint
[FK] product_id	bigint
[FK] variant_id	bigint?
qty	int
unit_price	decimal

coupons

[PK] id	bigint
[UQ] code	varchar(40)
type/value	enum/dec
min_order/max_disc	dec?
usage/per_user_limit	int?
starts/ends_at	ts?
is_active	bool

coupon_usages

[PK] id	bigint
[FK] coupon_id	bigint
[FK] user_id	bigint
[FK] order_id	bigint
discount	decimal
used_at	ts

Orders, money and fulfillment

Orders, money and fulfillment

orders

[PK] id	bigint
[UQ] order_no	varchar(20)
[FK] user_id	bigint?
status	enum
subtotal/disc/delivery/total	decimal
pay_method/status	enum
snapshot name/phone/addr	varchar/text
district/thana/area	varchar
[FK] address_id	bigint?
+2 more (see spec)	

order_items

[PK] id	bigint
[FK] order_id	bigint
vendor_id	bigint?
[FK] product/variant_id	bigint
name/sku snapshot	varchar
qty/price/total	int/dec

order_status_histories

[PK] id	bigint
[FK] order_id	bigint
from/to_status	enum
[FK] changed_by	bigint?
note	text?
created_at	ts

return_requests

[PK] id	bigint
[FK] order_id	bigint
[FK] order_item_id	bigint?
[FK] user_id	bigint
type	enum
reason	text
status	enum

payments

[PK] id	bigint
[FK] order_id	bigint
gateway	enum
[UQ] tran_id	varchar
amount/currency	dec/char
status	enum
payload	json?
paid_at	ts?

shipments

[PK] id	bigint
[FK] order_id	bigint
[FK] courier_id	bigint?
tracking_id	varchar?
charge/status	dec/enum
shipped/delivered_at	ts?

Campaigns, recovery and engagement

Campaigns, recovery and engagement

flash_sales

[PK] id	bigint
title_en/bn	varchar
starts/ends_at	ts
is_active	bool

flash_sale_products

[PK] id	bigint
[FK] flash_sale_id	bigint
[FK] product_id	bigint
sale_price	decimal
stock_cap/sold	int

abandoned_carts

[PK] id	bigint
[FK] cart_id	bigint
contact snapshot	varchar?
reminded_at	ts?
[FK] recovered_order_id	bigint?

reviews

[PK] id	bigint
[FK] user_id	bigint
[FK] product_id	bigint
[FK] order_id	bigint?
rating	tinyint
title/body	varchar?/text?
status	enum

wishlists

[PK] id	bigint
[FK] user_id	bigint
[FK] product_id	bigint
created_at	ts

notifications

[PK] id	bigint
[FK] user_id	bigint
type/title/body	varchar/text
data	json?
read_at	ts?

loyalty_transactions

[PK] id	bigint
[FK] user_id	bigint
[FK] order_id	bigint?
points	int
type/note	enum/text?
created_at	ts

referrals

[PK] id	bigint
[FK] referrer_id	bigint
[FK] referred_id	bigint
bonus/status	int/enum

product_views

[PK] id	bigint
user/session	bigint?/varchar?
[FK] product_id	bigint
viewed_at	ts

AI features and system tables

AI features and system tables

ai_generations

[PK] id	bigint
[FK] user_id	bigint
type	enum
input/output	text/json
provider/tokens	varchar?/int?
status	enum

chatbot_messages

[PK] id	bigint
user/session	bigint?/varchar?
role	enum
message	text
created_at	ts

settings

[PK] key	varchar(100)
value	json/text
group	varchar(40)

activity_logs

[PK] id	bigint
[FK] actor_id	bigint?
action/subject	varchar
ip/agent/meta	varchar?/json?
created_at	ts

backup_logs

[PK] id	bigint
[FK] started_by	bigint?
path/size/status	varchar?/enum
finished_at	ts?

3. Table specification (all columns)

users		
column	**type**	**notes**
id	bigint PK	auto
name	varchar(100)	required
phone	varchar(20) UQ+IDX	login + COD identity
email	varchar(190) UQ?	nullable, for receipts
password	hash	nullable for OTP-only? keep required
role	enum customer/admin/sup	RBAC gate
avatar	varchar?	media path
locale	char(2) default bn	bn/en/ar
loyalty_points	int default 0	cached balance
referral_code	varchar(20) UQ	earned on share
referred_by	FK users?	who invited
phone_verified_at	ts?	OTP verify
two_factor_secret	varchar?	admin 2FA
two_factor_enabled_at	ts?	admin 2FA
remember_token + timestamps	-	laravel default

addresses		
column	**type**	**notes**
id	bigint PK	
user_id	FK users cascade +IDX	
label	varchar(30)	Home/Office
recipient_name/phone	varchar	snapshot for rider
district/thana/area	varchar +IDX	charge lookup
street_address/postal_code	varchar?	
is_default	bool	one per user (app rule)

phone_otps		
column	**type**	**notes**
id	bigint PK	
phone	varchar(20)+IDX	
code_hash	varchar(255)	never plain
purpose	enum login/order/verify	
attempts	int default 0	max 5 then lock
expires_at	ts 5 min	
consumed_at	ts?	

categories		
column	**type**	**notes**
id	bigint PK	
parent_id	FK self?	2-level tree
name_en/name_bn/slug UQ	varchar	slug for /shop?cat=
icon/image	varchar?	emoji today, file later
sort_order/is_active	int/bool	

brands		
column	**type**	**notes**
id	bigint PK	
name/slug UQ	varchar	
logo/is_active	varchar?/bool	

products		
column	**type**	**notes**
id	bigint PK	
vendor_id	FK? nullable	NULL = own store; future marketplace
category_id/brand_id?	FK	+IDX
name/slug UQ	varchar	
subtitle	varchar?	short line under name
description_en/bn	mediumtext?	AI-assisted copy
price/old_price/cost	decimal(12,2)	margin = price-cost
sku UQ/barcode?	varchar	
stock_qty/low_stock_at	int	low-stock alert
status	enum draft/active/archived	
is_featured	bool	home row
rating_avg/reviews_count/sold_co	decimal/int	cached counters
seo_title/desc/keywords?	varchar?	
deleted_at?	softDeletes	keep order history safe

product_variants		
column	**type**	**notes**
id	bigint PK	
product_id	FK cascade +IDX	
name/sku UQ	varchar	Size/Color
price_override?	decimal?	null = base price
stock_qty	int	per-variant stock
attributes	json	{size,color}

product_media		
column	**type**	**notes**
id	bigint PK	
product_id/variant_id?	FK cascade	
type	enum image/video	
url	varchar(500)	storage path
sort_order/is_primary	int/bool	

delivery_zones		
column	**type**	**notes**
id	bigint PK	
district	varchar +IDX	
thana/area?	varchar?	null = whole district
charge	decimal(10,2)	auto charge at checkout
courier_id?	FK?	default courier
eta_days?	int?	
is_active	bool	

couriers		
column	**type**	**notes**
id	bigint PK	
name/code UQ	varchar	steadfast/redx/pathao...
api_config	json?	keys per env
is_active	bool	

carts		
column	**type**	**notes**
id	bigint PK	
user_id?	FK? +IDX	null = guest
session_id?	varchar?	guest key
coupon_id?	FK?	applied code

cart_items		
column	**type**	**notes**
id	bigint PK	
cart_id	FK cascade	
product/variant_id?	FK	
qty	int	
unit_price	decimal	snapshot at add
UQ(cart,product,variant)	unique	

coupons		
column	**type**	**notes**
id	bigint PK	
code UQ	varchar(40)	uppercased
type	enum percent/fixed/freesh	
value/min_order?/max_discount?	decimal?	
usage_limit?/per_user_limit?	int?	
starts/ends_at?/is_active	ts?/bool	

coupon_usages		
column	**type**	**notes**
id	bigint PK	
coupon/user/order_id	FK	UQ(order) one code per order
discount_amount	decimal	
used_at	ts	

flash_sales		
column	**type**	**notes**
id	bigint PK	
title_en/bn	varchar	
starts_at/ends_at/is_active	ts/bool	

flash_sale_products		
column	**type**	**notes**
id	bigint PK	
flash_sale_id/product_id	FK	UQ(pair)
sale_price	decimal	
stock_cap?/sold_qty	int	progress bar

orders		
column	**type**	**notes**
id	bigint PK	
order_no UQ	varchar(20)	CB-2026-000001
user_id?	FK? +IDX	null = guest
status +IDX	enum 7 states	placed>confirmed>processing>shipped>out_for_delivery>delivered (+cancelled)
subtotal/discount/delivery_charge	decimal	
payment_method	enum cod/bkash/nagad/ss	
payment_status	enum unpaid/paid/failed/re	
name/phone/address snapshot	stored copy	never rely on live address
district/thana/area + address_id?	varchar/FK?	

courier_id?/tracking_id?	FK?/varchar?	
coupon_id?	FK?	
notes?	text?	

order_items		
column	**type**	**notes**
id	bigint PK	
order_id	FK cascade +IDX	
vendor_id?	FK? nullable	future split-settlement
product/variant_id	FK (restrict)	keep history even if product edited
name/sku snapshot	varchar	
qty/unit_price/total	int/decimal	

order_status_histories		
column	**type**	**notes**
id	bigint PK	
order_id	FK cascade +IDX	
from/to_status	enum	
changed_by?	FK users?	
note?	text?	
created_at	ts	no updates

return_requests		
column	**type**	**notes**
id	bigint PK	
order/order_item/user_id	FK	
type	enum cancel/return/excha	
reason	text	
status	enum requested/approved	

payments		
column	**type**	**notes**
id	bigint PK	
order_id	FK +IDX	
gateway	enum	
tran_id UQ	varchar	gateway ref
amount/currency BDT	decimal/char(3)	
status	enum pending/paid/failed/	
payload?	json	IPN data
paid_at?	ts?	

shipments		
column	**type**	**notes**
id	bigint PK	
order_id	FK	
courier_id?	FK?	
tracking_id?	varchar? +IDX	customer tracking
charge/status	decimal/enum	
shipped/delivered_at?	ts?	
payload?	json	courier API reply

reviews		
column	**type**	**notes**
id	bigint PK	
user/product/order_id?	FK	order proves purchase
rating	tinyint 1-5	
title?/body?	varchar?/text?	
images?	json?	
status	enum pending/approved/re	moderation

wishlists		
column	**type**	**notes**
id	bigint PK	
user/product_id	FK	UQ(pair) +IDX user

notifications		
column	**type**	**notes**
id	bigint PK	
user_id	FK +IDX	
type	varchar (order/sale/system	
title/body	varchar/text (bn+en)	
data?	json	{order_id}
read_at?	ts?	

loyalty_transactions		
column	**type**	**notes**
id	bigint PK	
user/order_id?	FK	
points	int signed	+earn -redeem
type/note?	enum/text?	

referrals		
column	**type**	**notes**
id	bigint PK	
referrer_id	FK users	
referred_id	FK users UQ	one inviter
bonus_points/status	int/enum	

product_views		
column	**type**	**notes**
id	bigint PK	
user_id?/session_id?	FK?/varchar?	
product_id	FK +IDX	
viewed_at	ts +IDX	recently-viewed + recs

abandoned_carts		
column	**type**	**notes**
id	bigint PK	
cart_id	FK UQ	
contact snapshot?	varchar?	phone/email for SMS
reminded_at?	ts?	
recovered_order_id?	FK?	

ai_generations		
column	**type**	**notes**
id	bigint PK	
user_id	FK admin	
type	enum title/desc/seo/captio	
input_text?	text?	
output	mediumtext/json	
provider?/tokens?	varchar?/int?	cost log
status	enum	

chatbot_messages		
column	**type**	**notes**
id	bigint PK	
user_id?/session_id	FK?/varchar +IDX	
role	enum user/assistant	
message	text	

settings		
column	**type**	**notes**
key PK	varchar(100)	payments.pixel.analytics...
value	json	
group	varchar(40)	

activity_logs		
column	**type**	**notes**
id	bigint PK	
actor_id?	FK users?	
action	varchar +IDX	order.status, coupon.create...
subject	morph type+id	
ip/user_agent/meta?	varchar?/json?	fraud trail

backup_logs		
column	**type**	**notes**
id	bigint PK	
started_by?	FK users?	
path?/size?	varchar?/bigint?	
status	enum	
finished_at?	ts?	

4. Key relationships

- users 1-N addresses / phone_otps / orders / reviews / wishlists / notifications / loyalty / referrals / ai_generations
- categories self-tree 1-N; products N-1 categories/brands; variants + media N-1 products (cascade delete)
- carts 1-N cart_items; coupon N-1 carts (applied code)
- orders 1-N items / status_histories / payments / shipments; coupon N-1 orders; address snapshot copied, never joined for invoices
- flash_sales 1-N flash_sale_products N-1 products (UQ pair)
- reviews UQ(user,product,order?) - one verified review per purchase; wishlists UQ(user,product)
- abandoned_carts 1-1 carts; recovery links to recovered_order_id
- vendor_id NULLABLE on products + order_items = single-vendor today, marketplace later without schema change

Must-have indexes (besides PK/FK)

- users(phone), users(email), users(role); addresses(user_id, district); phone_otps(phone, expires_at)
- products(status, category_id), products(slug), variants(product_id), media(product_id)
- orders(order_no UQ, user_id, status, payment_status); items(order_id); payments(tran_id UQ, order_id)
- coupons(code UQ); usages(coupon_id, user_id); flash_sale_products(flash_sale_id, product_id)
- reviews(product_id, status); wishlists(user_id); notifications(user_id, read_at)
- product_views(product_id, viewed_at); shipments(tracking_id); activity_logs(action, created_at)

5. Backend build order (phases with exit criteria)

Phase 0 - foundation (week 1)

Build: Roles (customer/admin/support) on users, admin 2FA columns, settings + activity_logs, locale bn default, HTTPS + backups job.
Done when: Can log in as admin with 2FA; every admin write leaves an activity_log.

Phase 1 - catalog + inventory

Build: categories, brands, products, variants, media; pricing/discounts; stock + low-stock query; product policies (admin manage, public view active only).
Done when: Admin CRUD with images/videos + variants; storefront lists only active; low-stock list works.

Phase 2 - cart, checkout, money, delivery

Build: delivery_zones + couriers, carts/cart_items (guest via session), coupons + usages, orders + items + histories, payments (COD/bKash/Nagad/SSLCommerz), shipments, phone_otps; order-no generator; delivery-charge auto rule.
Done when: Guest can checkout with phone+address, charge auto-applies, COD order placed, online payment IPN marks paid, courier booking returns tracking id.

Phase 3 - post-order + account

Build: 7-state status machine + history, tracking page, return_requests, reviews (moderated), wishlists, addresses book, notifications, loyalty + referrals.
Done when: Status can only move forward legally; customer sees history, tracking, returns, points.

Phase 4 - marketing & analytics

Build: flash_sales + caps/sold counter (atomic increment), abandoned_carts detector + reminder job, product_views -> recently viewed + related/recommendations, settings keys for Pixel/Analytics + conversion events.
Done when: Flash sale respects cap and ends on time; abandoned SMS/email job recovers a test cart.

Phase 5 - AI features

Build: ai_generations log + provider adapter (queue jobs, fallback when provider down), chatbot_messages, sales-analysis snapshot job, voice-to-text upload endpoint.
Done when: Every AI output is logged with tokens; storefront works fully with AI provider disabled.

Phase 6 - hardening

Build: Rate-limit OTP/login/checkout, fraud rules (same phone X COD/day, RMA abuse), RBAC policies audit, backup verify/restore drill, SMS/email/WhatsApp confirmation templates (bn+en).
Done when: OTP brute-force blocked; restore drill from backup_logs passes; abuse order auto-flagged.

6. Cross-cutting rules

- Money in decimal(12,2), BDT; never float. Prices snapshotted on cart_items and order_items.
- Order address is a snapshot copy; address book edits must not rewrite old orders.
- Status changes only via service + history row; no direct status mass-assignment.
- Flash-sale sold_qty increment inside DB transaction with stock check (no oversell).
- All AI calls queued with timeout + graceful fallback text; log tokens for cost control.
- Files on storage disk, DB keeps paths only; product delete = soft delete.
- Feature tests per phase: checkout totals, coupon math, payment IPN, status machine, RMA flow, reminder job.